

Lecture Notes: Counting k -Dimensional Subspaces of an n -Dimensional Vector Space Using a Concrete Basis $v_1, \dots, v_n$

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1 Introduction

Let V be an n -dimensional vector space over a field K . We want to answer:

How many distinct k -dimensional subspaces does V have?

The answer depends on the field.

- If $K = \mathbf{F}_q$ is a finite field with q elements, then the number of k -dimensional subspaces is finite and is given by the *Gaussian binomial coefficient*.
- If K is infinite, then for $0 < k < n$ there are infinitely many k -dimensional subspaces.

In these notes we explain the finite-field formula in the most concrete way possible: we fix a basis

$$v_1, \dots, v_n$$

of V , write every vector in coordinates relative to that basis, and count subspaces by counting ordered bases inside V itself.

So the philosophy is this:

A k -dimensional subspace is determined by an ordered basis of k linearly independent vectors, and we can count these directly once a concrete basis $v_1, \dots, v_n$ is fixed.

2 Fixing a Concrete Basis $v_1, \dots, v_n$

From now on, assume that V is an n -dimensional vector space over $\mathbf{F}_q$, and fix a basis

$$v_1, \dots, v_n.$$

Then every vector $x \in V$ can be written uniquely in the form

$$x = a_1 v_1 + \dots + a_n v_n, \quad a_i \in \mathbf{F}_q.$$

Thus each vector of V corresponds uniquely to an n -tuple

$$(a_1, \dots, a_n) \in \mathbf{F}_q^n.$$

So although we do not identify V with $\mathbf{F}_q^n$ abstractly at first, the fixed basis $v_1, \dots, v_n$ lets us count vectors exactly as if we were in $\mathbf{F}_q^n$. In particular:

- V has exactly q^n elements;
- any r -dimensional subspace of V has exactly q^r elements.

Proposition 2.1. *Let $W \subseteq V$ be an r -dimensional subspace. Then W has exactly q^r elements.*

Proof. Choose a basis $w_1, \dots, w_r$ of W . Every vector in W has a unique expression

$$b_1 w_1 + \dots + b_r w_r, \quad b_i \in \mathbf{F}_q.$$

Since each coefficient has q possibilities, the total number of vectors in W is q^r . □

This simple fact is the engine behind the counting formula.

3 The Main Formula

Definition 3.1. For integers $0 \leq k \leq n$, the *Gaussian binomial coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})}.$$

Equivalently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

Theorem 3.2 (Main Theorem). *Let V be an n -dimensional vector space over $\mathbf{F}_q$, and fix a basis*

$$v_1, \dots, v_n.$$

Then the number of distinct k -dimensional subspaces of V is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

The proof will be carried out entirely inside V , using the fixed basis $v_1, \dots, v_n$ to count vectors and spans concretely.

4 The Counting Strategy

The argument has two steps.

1. Count all ordered k -tuples

$$(w_1, \dots, w_k)$$

of linearly independent vectors in V .

2. Divide by the number of ordered bases of one fixed k -dimensional subspace.

Why does this work? Because every ordered linearly independent k -tuple is an ordered basis of exactly one k -dimensional subspace, namely

$$\text{span}(w_1, \dots, w_k),$$

and every k -dimensional subspace has many ordered bases.

So the desired count is:

$$\#\{k\text{-dimensional subspaces of } V\} = \frac{\#\{\text{ordered linearly independent } k\text{-tuples in } V\}}{\#\{\text{ordered bases of one fixed } k\text{-dimensional subspace}\}}.$$

5 Counting Ordered Linearly Independent k -Tuples in V

We now count ordered tuples

$$(w_1, \dots, w_k) \in V^k$$

such that $w_1, \dots, w_k$ are linearly independent.

Because the basis $v_1, \dots, v_n$ is fixed, every vector in V has coordinates

$$w = a_1 v_1 + \cdots + a_n v_n.$$

Hence V has exactly q^n vectors.

Lemma 5.1. *The number of ordered linearly independent k -tuples in V is*

$$(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1}).$$

Proof. We choose the vectors one by one.

Step 1: choosing w_1

The first vector w_1 can be any nonzero vector of V . Since V has q^n vectors, there are

$$q^n - 1$$

choices for w_1 .

Step 2: choosing w_2

Once w_1 is fixed, the vector w_2 must not lie in $\text{span}(w_1)$. Since $\text{span}(w_1)$ is 1-dimensional, it contains exactly q vectors. Therefore the number of choices for w_2 is

$$q^n - q.$$

Step 3: choosing w_3

Once w_1, w_2 are linearly independent, the vector w_3 must not lie in

$$\text{span}(w_1, w_2).$$

This is a 2-dimensional subspace, so it contains exactly q^2 vectors. Hence the number of choices for w_3 is

$$q^n - q^2.$$

Continuing in this way, after choosing linearly independent vectors

$$w_1, \dots, w_{i-1},$$

their span has dimension $i - 1$, so it contains exactly q^{i-1} vectors. Thus the number of choices for w_i is

$$q^n - q^{i-1}.$$

Multiplying over $i = 1, \dots, k$, we obtain

$$(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1}).$$

□

Remark 5.2. This count depends only on the fact that V has a basis $v_1, \dots, v_n$. Once such a basis is fixed, the counting becomes completely concrete.

6 Counting Ordered Bases of One Fixed k -Dimensional Subspace

Now fix a k -dimensional subspace $W \subseteq V$. We want to count how many ordered bases W has.

Lemma 6.1. *A fixed k -dimensional subspace W has exactly*

$$(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})$$

ordered bases.

Proof. Since W has dimension k , choose a basis

$$u_1, \dots, u_k$$

of W . Then every vector of W has a unique expression

$$c_1 u_1 + \cdots + c_k u_k, \quad c_i \in \mathbf{F}_q.$$

So W has exactly q^k vectors.

Now count ordered bases of W exactly as before:

- the first basis vector can be any nonzero vector of W : $q^k - 1$ choices;
- the second must lie outside the span of the first: $q^k - q$ choices;
- the third must lie outside the span of the first two: $q^k - q^2$ choices;
- and so on.

Thus the number of ordered bases of W is

$$(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1}).$$

□

7 Proof of the Main Theorem

We now combine the two counts.

Proof of the Main Theorem. Every ordered linearly independent k -tuple

$$(w_1, \dots, w_k)$$

in V spans a unique k -dimensional subspace

$$\text{span}(w_1, \dots, w_k).$$

Conversely, every k -dimensional subspace contributes exactly as many such ordered tuples as it has ordered bases, namely

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1}).$$

Therefore,

$$\#\{k\text{-dimensional subspaces of } V\} = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}.$$

By definition, this is exactly

$$\begin{bmatrix} n \\ k \end{bmatrix}_q.$$

□

8 What the Basis $v_1, \dots, v_n$ Is Doing

It is worth emphasizing the role of the concrete basis

$$v_1, \dots, v_n.$$

We are not saying that every k -dimensional subspace is simply

$$\text{span}(v_{i_1}, \dots, v_{i_k})$$

for some choice of indices $i_1 < \dots < i_k$. That would count only the coordinate subspaces, and there are many more subspaces than that.

Rather, the basis $v_1, \dots, v_n$ is used to do three things:

1. to write every vector of V uniquely in coordinates;
2. to conclude that V has exactly q^n vectors;
3. to conclude that each r -dimensional subspace has exactly q^r vectors.

Once these facts are known, the counting argument becomes possible.

Remark 8.1. So the basis $v_1, \dots, v_n$ is a counting device, not a list from which all subspaces are built by merely selecting subsets of basis vectors.

9 Coordinate Subspaces Versus All Subspaces

To avoid confusion, let us state this explicitly.

Definition 9.1. A *coordinate k -subspace* relative to the basis $v_1, \dots, v_n$ is a subspace of the form

$$\text{span}(v_{i_1}, \dots, v_{i_k})$$

for some

$$1 \leq i_1 < \dots < i_k \leq n.$$

There are exactly

$$\binom{n}{k}$$

such coordinate k -subspaces.

However, in general,

$$\binom{n}{k} \neq \left[\begin{matrix} n \\ k \end{matrix} \right]_q.$$

The Gaussian coefficient is usually much larger.

Example 9.2. In $\mathbf{F}_2^3$, the number of coordinate 1-dimensional subspaces relative to a fixed basis v_1, v_2, v_3 is only

$$\binom{3}{1} = 3,$$

namely

$$\text{span}(v_1), \quad \text{span}(v_2), \quad \text{span}(v_3).$$

But the total number of 1-dimensional subspaces is

$$\left[\begin{matrix} 3 \\ 1 \end{matrix} \right]_q = \frac{2^3 - 1}{2 - 1} = 7.$$

So four more lines appear, such as

$$\text{span}(v_1 + v_2), \quad \text{span}(v_1 + v_3), \quad \text{span}(v_2 + v_3), \quad \text{span}(v_1 + v_2 + v_3).$$

This shows clearly why ordinary binomial coefficients are not the answer here.

10 Examples

Example 10.1 (Lines in V). The number of 1-dimensional subspaces of V is

$$\left[\begin{matrix} n \\ 1 \end{matrix} \right]_q = \frac{q^n - 1}{q - 1}.$$

Indeed, a 1-dimensional subspace is determined by any nonzero vector, but each such subspace has exactly $q - 1$ nonzero vectors. So

$$\#\{\text{lines in } V\} = \frac{q^n - 1}{q - 1}.$$

Example 10.2 (Hyperplanes in V). The number of $(n - 1)$ -dimensional subspaces of V is

$$\left[\begin{matrix} n \\ n - 1 \end{matrix} \right]_q = \left[\begin{matrix} n \\ 1 \end{matrix} \right]_q = \frac{q^n - 1}{q - 1}.$$

Example 10.3 (The case $V = \mathbf{F}_2^2$ with basis v_1, v_2). The vectors of V are

$$0, \quad v_1, \quad v_2, \quad v_1 + v_2.$$

So the nonzero vectors are

$$v_1, \quad v_2, \quad v_1 + v_2.$$

Each 1-dimensional subspace contains exactly one nonzero vector because the only nonzero scalar in $\mathbf{F}_2$ is 1. Thus the three 1-dimensional subspaces are

$$\text{span}(v_1), \quad \text{span}(v_2), \quad \text{span}(v_1 + v_2).$$

Hence

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = \frac{2^2 - 1}{2 - 1} = 3.$$

Example 10.4 (The case $V = \mathbf{F}_2^3$ with basis v_1, v_2, v_3). The formula gives

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}_q = \frac{(2^3 - 1)(2^3 - 2)}{(2^2 - 1)(2^2 - 2)} = \frac{7 \cdot 6}{3 \cdot 2} = 7.$$

So V has exactly seven 2-dimensional subspaces.

11 A Fully Concrete Example Using v_1, v_2, v_3

Let V be a 3-dimensional vector space over $\mathbf{F}_2$ with fixed basis

$$v_1, v_2, v_3.$$

Then the vectors of V are

$$0, \quad v_1, \quad v_2, \quad v_3, \quad v_1 + v_2, \quad v_1 + v_3, \quad v_2 + v_3, \quad v_1 + v_2 + v_3.$$

Let us count the 2-dimensional subspaces directly.

Step 1: count ordered bases of 2-dimensional subspaces

To choose an ordered basis (w_1, w_2) :

- w_1 can be any nonzero vector: 7 choices;
- w_2 must not lie in $\text{span}(w_1)$. Since $\text{span}(w_1)$ has exactly 2 elements, namely 0 and w_1 , there are 6 choices.

So there are

$$7 \cdot 6 = 42$$

ordered bases of 2-dimensional subspaces.

Step 2: count ordered bases of one fixed 2-dimensional subspace

Take for example

$$W = \text{span}(v_1, v_2).$$

Its vectors are

$$0, \quad v_1, \quad v_2, \quad v_1 + v_2.$$

Thus its nonzero vectors are

$$v_1, \quad v_2, \quad v_1 + v_2.$$

To choose an ordered basis of W :

- the first basis vector has 3 choices;
- the second must be different from the first and nonzero, so it has 2 choices.

Hence W has

$$3 \cdot 2 = 6$$

ordered bases.

Step 3: divide

Therefore the number of distinct 2-dimensional subspaces is

$$\frac{42}{6} = 7.$$

This matches the formula

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}_q = 7.$$

12 The General Linear Group and Ordered Bases

The denominator in the Gaussian coefficient is also the size of the general linear group.

Proposition 12.1. *The number of ordered bases of a k -dimensional vector space over $\mathbf{F}_q$ is*

$$|\text{GL}_k(\mathbf{F}_q)| = (q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1}).$$

Proof. An invertible $k \times k$ matrix over $\mathbf{F}_q$ is exactly an ordered basis of $\mathbf{F}_q^k$, written as its columns. Counting such ordered bases gives the formula. $\square$

So we may rewrite the counting theorem as

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{\text{number of ordered linearly independent } k\text{-tuples in } V}{|\text{GL}_k(\mathbf{F}_q)|}.$$

13 Special Cases

Proposition 13.1. *For every n and q ,*

$$\begin{aligned}\begin{bmatrix} n \\ 0 \end{bmatrix}_q &= 1, \\ \begin{bmatrix} n \\ 1 \end{bmatrix}_q &= \frac{q^n - 1}{q - 1} = 1 + q + q^2 + \cdots + q^{n-1}, \\ \begin{bmatrix} n \\ n-1 \end{bmatrix}_q &= \begin{bmatrix} n \\ 1 \end{bmatrix}_q, \\ \begin{bmatrix} n \\ n \end{bmatrix}_q &= 1.\end{aligned}$$

Proof. The cases $k = 0$ and $k = n$ are immediate. The formula for $k = 1$ follows by counting nonzero vectors and dividing by the $q - 1$ nonzero scalar multiples of each one. The case $k = n - 1$ follows from symmetry. $\square$

14 Symmetry

Proposition 14.1. *For $0 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q.$$

Proof. This follows from the standard product formulas for the Gaussian binomial coefficient. $\square$

Remark 14.2. This symmetry says that the number of k -dimensional subspaces equals the number of $(n - k)$ -dimensional subspaces.

15 Another Form of the Formula

The Gaussian coefficient may also be written as

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \cdots (q - 1)}.$$

This form is often convenient for algebraic manipulation and makes the symmetry more visible.

16 What Happens Over Infinite Fields?

If the base field is infinite, then the answer is usually infinite.

Proposition 16.1. *Let V be an n -dimensional vector space over an infinite field K . If $0 < k < n$, then V has infinitely many distinct k -dimensional subspaces.*

Proof. It suffices to consider $k = 1$. Let u_1, u_2 be linearly independent vectors in V . For each scalar $a \in K$, the vector

$$u_1 + au_2$$

spans a 1-dimensional subspace

$$\text{span}(u_1 + au_2).$$

Different values of a give different lines. Since K is infinite, there are infinitely many such lines. Hence there are infinitely many 1-dimensional subspaces, and therefore infinitely many k -dimensional subspaces for $0 < k < n$. $\square$

17 Grassmannians

Definition 17.1. The set of all k -dimensional subspaces of an n -dimensional vector space is called the *Grassmannian*, denoted

$$\text{Gr}(k, n).$$

Over $\mathbf{F}_q$, the Grassmannian is a finite set and

$$|\text{Gr}(k, n)| = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

So the Gaussian binomial coefficient counts the points of the Grassmannian over $\mathbf{F}_q$.

18 Conceptual Summary

Let us summarize the whole argument in the concrete basis language.

Fix a basis

$$v_1, \dots, v_n$$

of the n -dimensional vector space V over $\mathbf{F}_q$.

1. Every vector of V has unique coordinates relative to $v_1, \dots, v_n$, so V has q^n elements.
2. Every r -dimensional subspace has q^r elements.
3. Ordered linearly independent k -tuples in V can be counted as

$$(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1}).$$

4. One fixed k -dimensional subspace has

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})$$

ordered bases.

5. Dividing gives the number of distinct k -dimensional subspaces.

Thus

$$\#\{k\text{-dimensional subspaces of } V\} = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})} = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

19 Final Theorem

Theorem 19.1 (Final Form). *Let V be an n -dimensional vector space over $\mathbf{F}_q$, and fix a concrete basis*

$$v_1, \dots, v_n.$$

Then the number of distinct k -dimensional subspaces of V is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

The proof is obtained by counting ordered bases of k -dimensional subspaces relative to the fixed basis $v_1, \dots, v_n$.

20 Exercises

Exercise 20.1. Let V be a 3-dimensional vector space over $\mathbf{F}_2$ with basis v_1, v_2, v_3 . List all vectors of V .

Exercise 20.2. Show directly that

$$\text{span}(v_1 + v_2) \neq \text{span}(v_1)$$

in a vector space over $\mathbf{F}_2$.

Exercise 20.3. Using the basis v_1, v_2, v_3 , count the number of 1-dimensional subspaces of V when V is over $\mathbf{F}_2$.

Exercise 20.4. Prove that an r -dimensional subspace over $\mathbf{F}_q$ has exactly q^r elements.

Exercise 20.5. Compute $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$ when $q = 2$.

Exercise 20.6. Compute $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$ when $q = 3$.

Exercise 20.7. Explain why choosing k vectors from the fixed basis $v_1, \dots, v_n$ counts only coordinate subspaces, not all k -dimensional subspaces.

Exercise 20.8. Show that the number of coordinate k -subspaces relative to $v_1, \dots, v_n$ is $\binom{n}{k}$.

21 Selected Answers

- For $q = 2$,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(2^4 - 1)(2^4 - 2)}{(2^2 - 1)(2^2 - 2)} = \frac{15 \cdot 14}{3 \cdot 2} = 35.$$

- For $q = 3$,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(3^4 - 1)(3^4 - 3)}{(3^2 - 1)(3^2 - 3)} = \frac{80 \cdot 78}{8 \cdot 6} = 130.$$

- The number of coordinate k -subspaces relative to $v_1, \dots, v_n$ is

$$\binom{n}{k}.$$

One-Sentence Takeaway

Once a concrete basis $v_1, \dots, v_n$ is fixed, vectors and subspaces can be counted explicitly, and the number of distinct k -dimensional subspaces of an n -dimensional vector space over $\mathbf{F}_q$ is the Gaussian binomial coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$.